

EVOTE

Name: Jake Nagle

Student ID: W20109020

LECTURER: COLM DUNPHY

SUPERVISOR: CATHERINE FITZPATRICK

MODULE: PROJECT

COURSE: Higher Diploma in Computer Science

Declaration

I declare that the work which follows is my own, and that any quotations from any sources (e.g. books, journals, the internet) are clearly identified as such by the use of ‘single quotation marks’, for shorter excerpt and identified italics for longer quotations. All quotations and paraphrases are accompanied by (date, author) in the text and a fuller citation is the bibliography. I have not submitted the work represented in this report in any other course of study leading to an academic award.

Student..... Date

Work Place Mentor..... Date

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Proposal / Objectives

Project Name:

EVOTE - An imaginary secure, user friendly electronic voting system for election polling implemented as a full stack application addressing UI for disabilities (accessibility), spoiled votes, including a dashboard for results.

Project Stack Area:

Full-Stack Web Application

Introduction / Background

In Ireland's most recent presidential election, national voter turnout was 45.83%, compared to 43.87% in the 2018 campaign. Although turnout increased slightly in 2025, it is still worryingly low. A study by Maynooth University identified several key factors associated with lower voter turnout (Kavanagh and Durkan, n.d.), including:

- High population mobility
- Urban-based employment patterns
- Distance from polling stations
- Voting held on weekdays

This project takes these findings as a starting point, exploring how technology could address such barriers and encourage greater participation through a secure, accessible, and user-friendly online voting system.

This project is strictly a prototype and is intended solely for technical experimentation and demonstration purposes. It does not collect, process, store, or interact with any real or personal data. All data used within the project is entirely fictional or simulated data created for development and testing only.

Project Objectives / Aims

The goal of this project is to design and implement a secure web-based voting application that allows registered fictitious users to cast their votes online.

Specific Objectives Include:

- Implement robust security features to ensure confidentiality, integrity, and fairness of votes.
- Enable fictitious users to register and authenticate securely.
- Provide an intuitive voting interface where fictitious users can select candidates.
- Display a voting summary dashboard, including interactive maps showing vote distribution by county.
- Maintain a historical record of previous elections and voting patterns.

Other Purposed Features for the Web Application

- Allow voters to submit vote just once.
- Secure enough to only allows eligible voters to vote i.e. registered voters.
- Allow voters to review their vote before submission reducing potential for human error.
- Has a security and protection element that will protect against cyberattacks.

Motivation

The recent 2025 presidential election sparked my interest in how technology could enhance Ireland's electoral process. After reviewing turnout data, I became curious about whether digital solutions could make voting more accessible and convenient, potentially increasing participation.

From a personal perspective, this project gives me the opportunity to apply and strengthen my full-stack development skills. It allows me to build a complete front-end and back-end application that will make a strong addition to my GitHub portfolio.

Problems the Project Intends to Address

- Heavy reliance on paper-based systems.
- Inaccessibility for Irish residents who are abroad or travelling on voting day.
- Time-consuming manual vote counting.
- Limited accessibility for people with mobility or travel difficulties.
- Giving the opportunity to vote to those who cannot/do not want to leave their house the day of the election (e.g. due to family duties or plans on the day of the elections).
- Lack of convenience and modernising the voting process.

Potential Challenges / Risks

1. **Security:** Ensuring data confidentiality, vote integrity, and system resistance to cyberattacks.
2. **Ethical and Legal Considerations:** Obtaining ethics approval and addressing data protection concerns related to online voting.
3. **Scalability:** Maintaining reliable performance when multiple fictitious users access the system at the same time.
4. **Clarity on results:** Ensuring that auditing of results is possible.

RAMP Phase – Research, Analysis, Modelling & Planning (Analysis & Design Phase)

Research / Analysis

Requirements analysis

The web application will be mainly focused on access to the fictional voting for people that cannot travel to the polling stations on the day required. These people could include a variety such as people who have difficulties travelling on the day of voting, Irish citizens who are living abroad so, people with disabilities who may need assistance to get to the arranged voting location, or people with responsibilities that require them to stay home bound for the entire day. The goal is to create a site that users could visit to cast their vote in order to eliminate the need for travel, potentially increasing the voting numbers.

To allow the web application to run it would need to implement multiple features. On arrival to the web application a voter would need to be able have a register and login method.

Voting would fall into the category of a sensitive area, therefore there would be a need for a high level of authentication on the application.

Upon registration a voter should be able to login and view a landing page. Irelands previous presidency vote has a dashboard with a summary of all the voting by number with a mapping of Ireland with the voting of each county summarised and this may be a good layout to follow for the site landing page.

What page to visit to cast a vote should be made clear for those who do not use technology often. There should be a page with all voting candidates clearly shown and a feature enabled where they can click the picture or box of the candidate to vote. A pop-up yes or no feature should be enabled to double check whether a user wants to cast their vote for that candidate. Confirmation on page on who they have voted for.

Main features to implement within the code:

- A user can only vote once.
- Authentication of user.
- Review of vote before submission.
- Increase the total number of votes once the user has voted.
- Increase the total number of votes for a candidate once the user has voted for a candidate.
- Increase the numbers of voting based on county.
- An un-registered user does not have access to the voting page.
- A user should be able to view the candidates they can vote for.

Nice to have features:

- Mapping of Ireland.
- Vote count by County.

Accessibility

Initially my goal was to build a web application due to the lack of accessibility to a voting centres or not having the ability to drive. After looking into how I could improve this aspect, I started I began to realise that accessibility goes a lot deeper than not having the ability to drive. I came across different articles which elaborated accessibility also meant having conditions where reading text on a screen can be incredibly difficult. With this in mind, I decided to expand my exploration on accessibility and see how I could format the layout of my site to improve it in general and for individuals who struggle with conditions that affect their ability to read text on a screen. The two conditions I focused on were dyslexia and colour blindness.

For lots of people a simple action such as reading text on the computer can be a challenge. Written syntax can be difficult for people with dyslexia. Dyslexia is disability which can cause the affected to mis-read words or have to read entire sentences slowly with many inaccuracies. Worldwide, dyslexia affects 1 in 10 people (Dyslexia Ireland, 2025).

Another group with accessibility problems are people with colour blindness. A person with colour blindness cannot differentiate between two different colours. Colour blindness affects 1 in 12 men and one in every 100 women (Kenny, 2018).

The accessibility framework proposed by Brickfield focuses on developing practical skills for creating digital content that will be accessible to everybody. The framework highlights multiple key design areas including images, media, tables, links, text, and layout. Brickfield provides guidance on what to apply in each design area to make content more accessible. By following these principles, developers can create web content that is easier to read, navigate, and understand for those with disabilities, including those with dyslexia and colour blindness (Snow, 2025).

How to build the web application to suit accessibility

Dyslexia

As dyslexia is a barrier to a person's ability to read, I will implement the following features to aid with their condition:

- All written text must be in simple, plain language to reduce the reading
- Structure content using headings, bullet points, and clear page titles
- The content must be well structured by using clear headings, lists and short paragraphs
- Use consistent layout and structured page organisation
- The content must have an increased line spacing of at least 1.5
- The content must have increased word and letter spacing
- The web application must allow a user to change the size of the UI, increasing the size of the text without it having an effect on the application performance or functionality
- Avoid excessive capital letters which can make text harder to read
- The web application will have clear links for navigation and descriptive labels that are easy to understand
- The web application will use text alternatives so text can be read out

Colour blindness

Most people with colour blindness have problems telling the red and green colours apart. This is the most common type (Kenny, 2018). Therefore, to improve the accessibility of my web application for people with colour blindness condition I will ensure:

- Error messages will be displayed in black text
- Additional indicators such as icons, text labels, or patterns used to ensure that information remains understandable
- Background colours set in style sheets will not be use red or green colour
- Good contrast in colour between text and background colours
- Use clear link styling such as underlining text so links are easily found

Project Management

For the development of my project, I divided my work over 6-week sprints. In order to manage where I spend my time from day to day and in what area of work I focus on, I used the web application Trello. Trello is an application that allows the user to add cards which all have text boxes. The text boxes can then be used to document each piece of work done or that needs to carry out on a later date. The application also has check boxes to be ticked when work is completed. I used this to manage my project on a daily and weekly basis. I created a card for each weekly sprint and added the work that needed to be done onto to keep track of the project details. Once initially setup I shared my Trello platform with my supervisor who could also view the application.

Iteration 1 – Install Frameworks / languages

On week iteration I downloaded Visual Studio and SQL Server Microsoft Management from the official Microsoft websites. I then went onto GitHub and created a repository for my final year project. Once completed, I moved on to Visual Studio Code where I installed Angular and created my first branch frontend. I quickly moved to backend where I created a .NET project onto Visual Studio, installed all the packages needed for project, and created another branch specifically for backend work.

Iteration 2 – Create Database

On the second iteration I focused on setting up the database. I created data models using the ER diagram. I configured a connection between the SQL Server and Visual Studio within the root folder to ensure that data migrates into the database and the database gets created using the data models. Next, using PowerShell CLI I migrated and updated the SQL Server to now display tables, ready for data to be inputted into them.

Iteration 3 – Frontend Landing Page / First API Creation

It was on iteration 3 where I decided to showcase work client side. I developed pages frontend to show a home page that displayed a site header that allowed for the user to navigate from the application using. I also added register and login buttons that lead to basic login and register pages with forms.

Backend I began to create view models to as schemas for what would be displayed on page and to use with the development of my API. The API was to be developed fully backend first using services, sent out using a controller and called from the frontend using services. I performed this cycle for user details, and later for vote, auth and test data.

Iteration 4 – Security

As my project was based on a sensitive topic it needed to have security aspects. For this I developed password hashing and the creation of a JWT token. I installed a BCrypt package backend to perform the password hashing. For the JWT token, I added helper method to create the token and then use that method to generate a token for the user after they register.

Iteration 5 – Testing

On iteration 5 after my services, controllers, view modes, and data models were created I went about testing my API methods to ensure they worked correctly before moving to frontend. I did use Swagger and mock data of users. I corrected any issues with the API calls. I also added a MSTest project to my files which I intended to use to add all API calls to as well to ensure that they worked correctly.

My idea was that I could use mock data or databases, and I did add some tests from the API calls into the test file. I tested the register on the frontend UI and seen that the database table got filled with the user details meaning all 3 frameworks were interacting with each other.

Iteration 6 – API (Frontend)

The API calls were created backend now I needed to grab them frontend. I did this using services specifically Observables and dependency injection. All API calls were created in a service file for example, auth service. Like a fetch which return a promise an Observable returns a subscribe and must be used to retrieve the data. In each component I have initialised a view model, subscribed to an API call from a service and used the method within the HTML template. I added two-way bindings, validation using required, styling using CSS, and interpolation in the HTML as well.

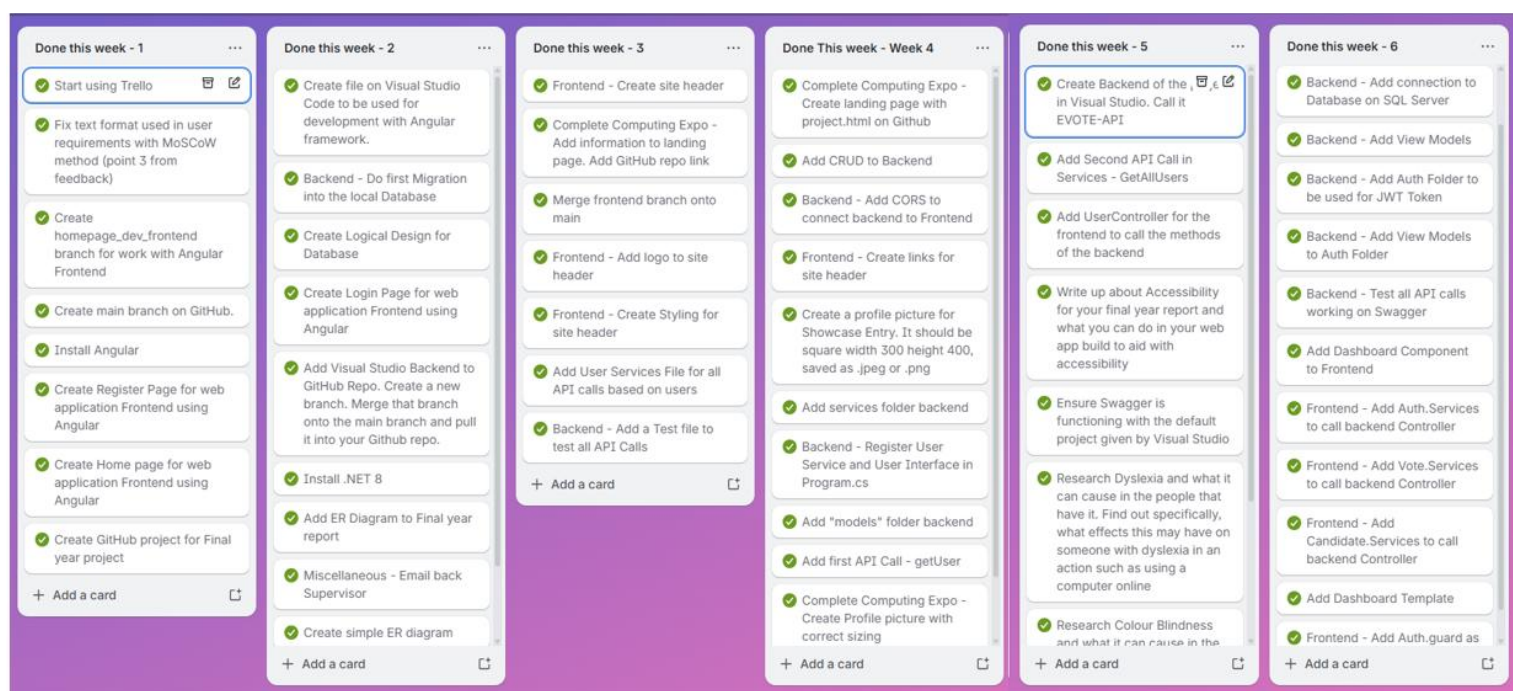


Figure 1: Trello Board

Analysis – Project Purpose

The purpose of this project is to develop a web application that will it have a front-end, backend and database framework working cohesively. My main focus will be on users who are unable to travel or have a disability and from a Computer Science viewpoint, building the application itself. While there will be elements of security applied this is not intended to be a security project. Lastly, I have the chosen the project area as a method of how a person may implement electronic voting in a presidential election campaign. My intention is not to showcase a project from a political standpoint but to research and display the technologies and frameworks that could be used in such a scenario.

Requirements for the System

To describe the Requirements for the System implemented I will use the MosCow method. This method will help prioritize the important features and break down exactly what is needed by detailing what the system must have, should have, could have and won't have (Visual Paradigm, 2023).

Term	Definitions
Must Have	Mandatory requirements essential for the project
Should Have	Important requirements but not critical to the project
Could Have	Requirements prioritized based on time left to complete the project
Won't Have	Requirements that will have no bearing on the successful outcome of the project

Table 1: MosCow method

1. Must Have

- i. The web application must allow users vote for a candidate.
- ii. The web application must allow a user to login to a dashboard page using a registration number.
- iii. The web application must allow a user to register to vote.
- iv. The web application must provide a registration number automatically upon registering for to vote.
- v. The web application must allow for viewing of all registered user's details on an analysis page.
- vi. The web application must have a method for counting the votes automatically.
- vii. The web application must allow for a summary of the voting to be seen on the analysis page of the admin.
- viii. The web application must allow for different user roles including standard user and an admin.
- ix. The web application must have functional tests by a user and testing of API methods to demonstrate how different sections of the application work.
- x. The web application must have security implementations including JWT tokens, sessions and cookies.

2. Should Have

- i. The web application should have a further higher level of security through the use of password hashing.
- ii. The web application should not allow the user to vote for again once they have voted for a candidate.
- iii. The web application should confirm with the user they have selected their desired candidate before submitting a vote to avoid human error.

3. Could Have

- i. The web application could filter the users table on the analysis page of an admin, filtering the voting of an individual heading of the table.
- ii. The web application could use an API key with Google Authentication to help secure the application.

4. Won't Have

- i. The application will not real user details to test any of the feature and instead use fictious people create by AI.

Methodologies

There are three different type of methodologies that a software development team can implement. The three main approaches are:

1. Waterfall
2. V-Shape
3. Agile

Waterfall is a sequential and linear approach where each phase must be completed before moving onto the next. The V-Shape Model in the SDLC is a parallel development and testing approach (GeeksforGeeks, 2023).

Agile is a SDLC methodology which focuses on the short quick bursts of development. It is an iterative approach to achieve continuous improvement and continuous development. Agile allows for responding to the changing of requirements to be handled easily due to its shorter cycles. In a real-world scenario it allows for continuous communication with a customer. The use an Agile approach leads to quicker feedback, better team collaboration, and in general more successful project outcomes (Atlassian, 2025).

Having looked at the methodologies, I will take the Agile approach to my Software Development Life Cycle as it puts a priority the higher value work, ensures code changes are integrated and tested frequently, and allows for continuous improvement (GeeksforGeeks, 2023).

Main Frameworks

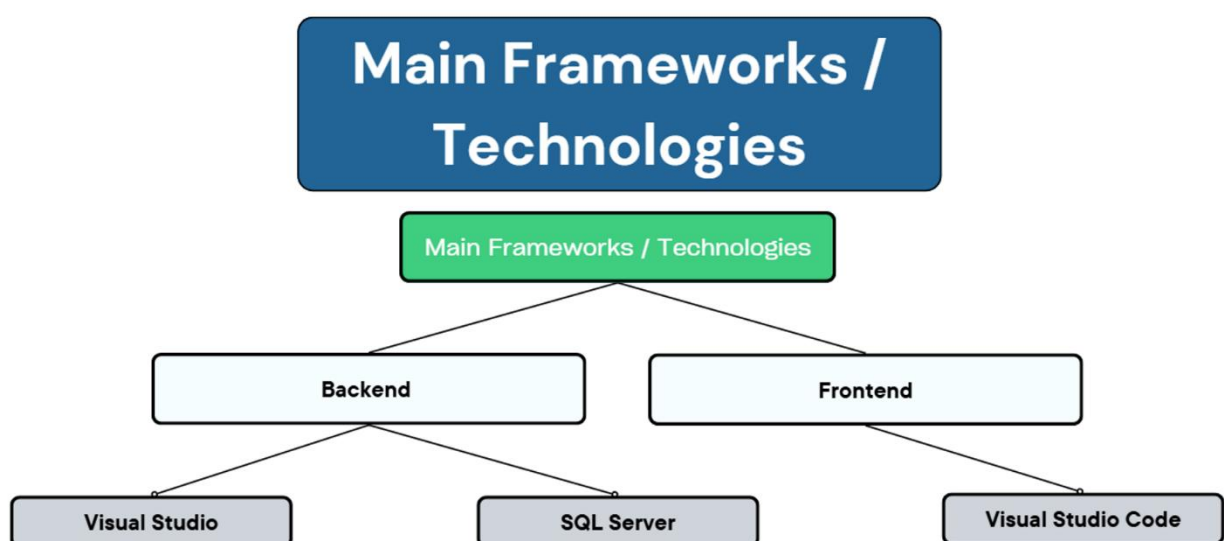


Figure 2: Frameworks

What is Visual Studio?

Visual Studio is an integrated development environment (IDE) that can be used to develop an entire application. It is a powerful tool for developers who can use it to write, edit, debug, and build code, and then deploy an app. Visual Studio includes features such as compilers, code completion tools, source control, extensions, and more to develop an application at every stage of software development life cycle (anandmeg, 2025).

Comparison with other IDEs

	Visual Studio	IntelliJ IDEA
User Interface	Cluttered	Cleaner and more organised
Language Support	Supports a wide range of languages including C, C# and JavaScript	Preferred choice for Java
Code Completion and Navigation	Provides intelligent suggestions, auto-completion and quick navigation	Provides intelligent suggestions, auto-completion and quick navigation
Build and Deployment Tools	More advanced build and deployment tools. Easier to manage in Microsoft eco-system	Would require more configurations to specific platforms
Plugin Ecosystem	Less plugins available	Robust third-party plugins
Pricing and Licensing	Free & paid license available	Free & paid license available

Table 2: IDE Comparisons

Summary:

IntelliJ IDEA is a very powerful tool and would certainly be the choice if I were developing a project in Java. Visual studio is much more flexible with the languages that can be used on the framework, provides the ability for advanced build and deployment. It also integrates with other Microsoft technology seamlessly (StackShare, 2025).

Connected Frameworks

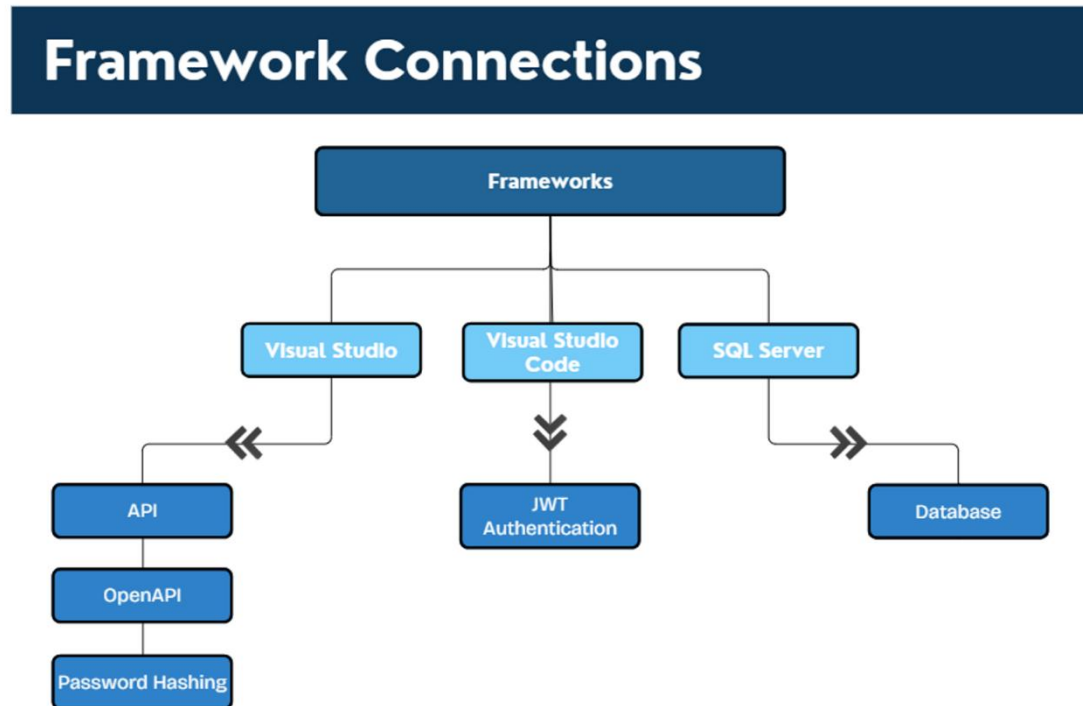


Figure 3: Framework connections

Tech Stack Selected for Project

Tech Stack			
	Frontend	Backend	Database
Framework	Visual Studio Code	Visual Studio	SQL
Language 1	Angular	C#	
Language 2	Typescript	.NET	
Language 3	HTML		

Figure 4: Tech Stack

Security

Feature	Produces	Test
JWT	User tokens	The return of a successful token will be tested for a on the www.jwt.io official site. As the token will be used in combination with API calls, the platform Swagger will also be used.
BCrypt	Hashed Password – Unable to read	The hashed password will be added to the user details table within the Database when a user registers

Table 3: Security features

Modelling

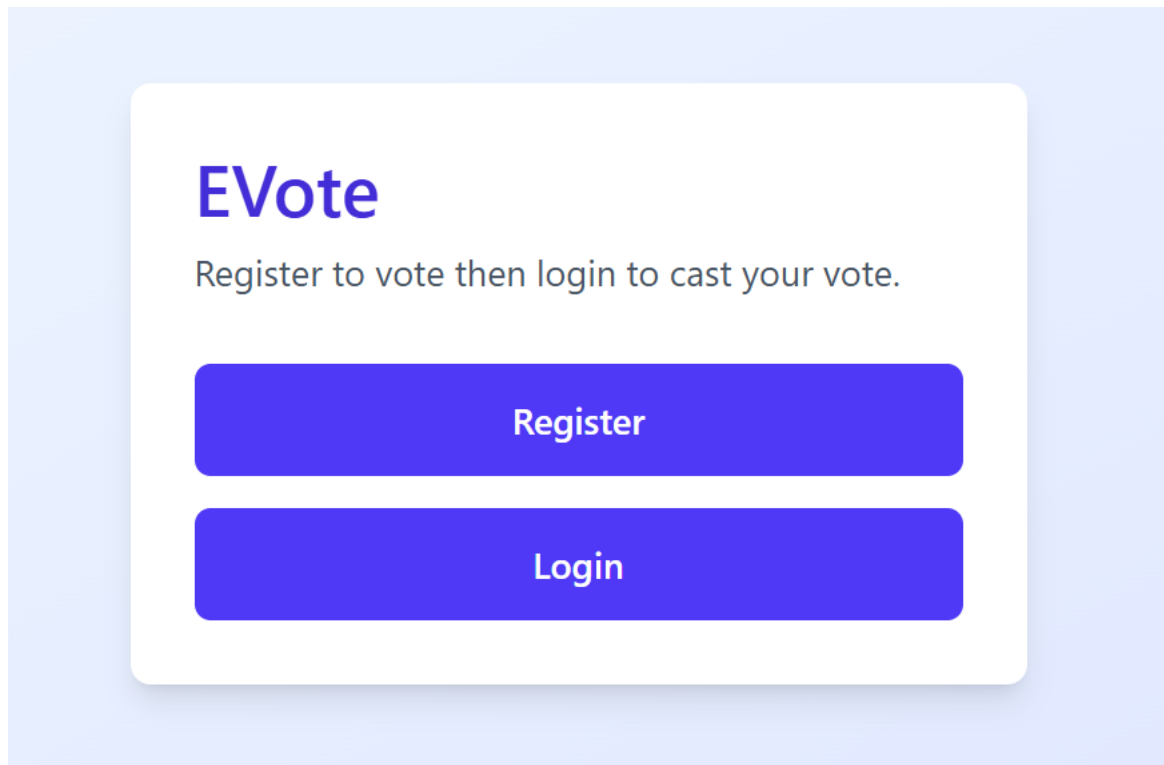


Figure 5: Evote home page

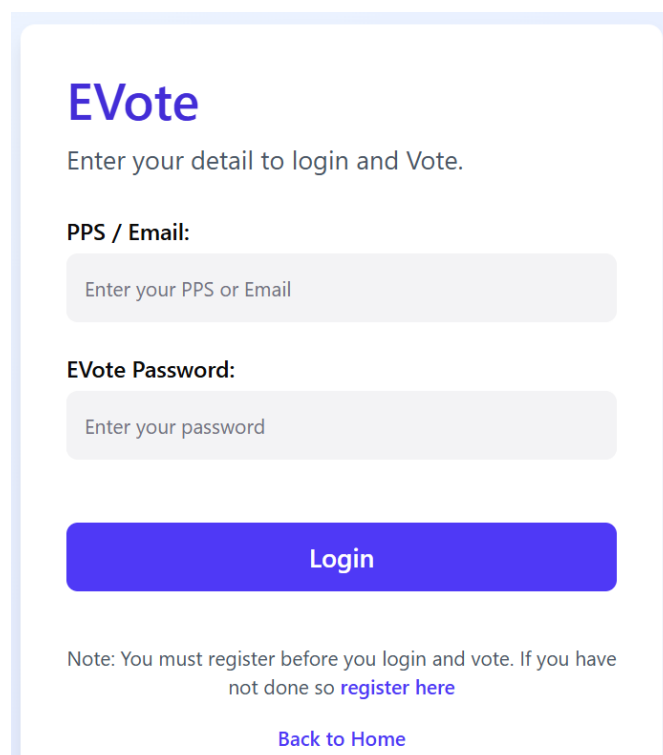


Figure 6: Evote figma Register page

EVote

Register Page

Name:

PPS:

Email:

Password for EVote:

Phone:

County:

DOB:

[Register](#)

[Back to Home](#)

Figure 7: Creation of Register page of Web Application on Figma

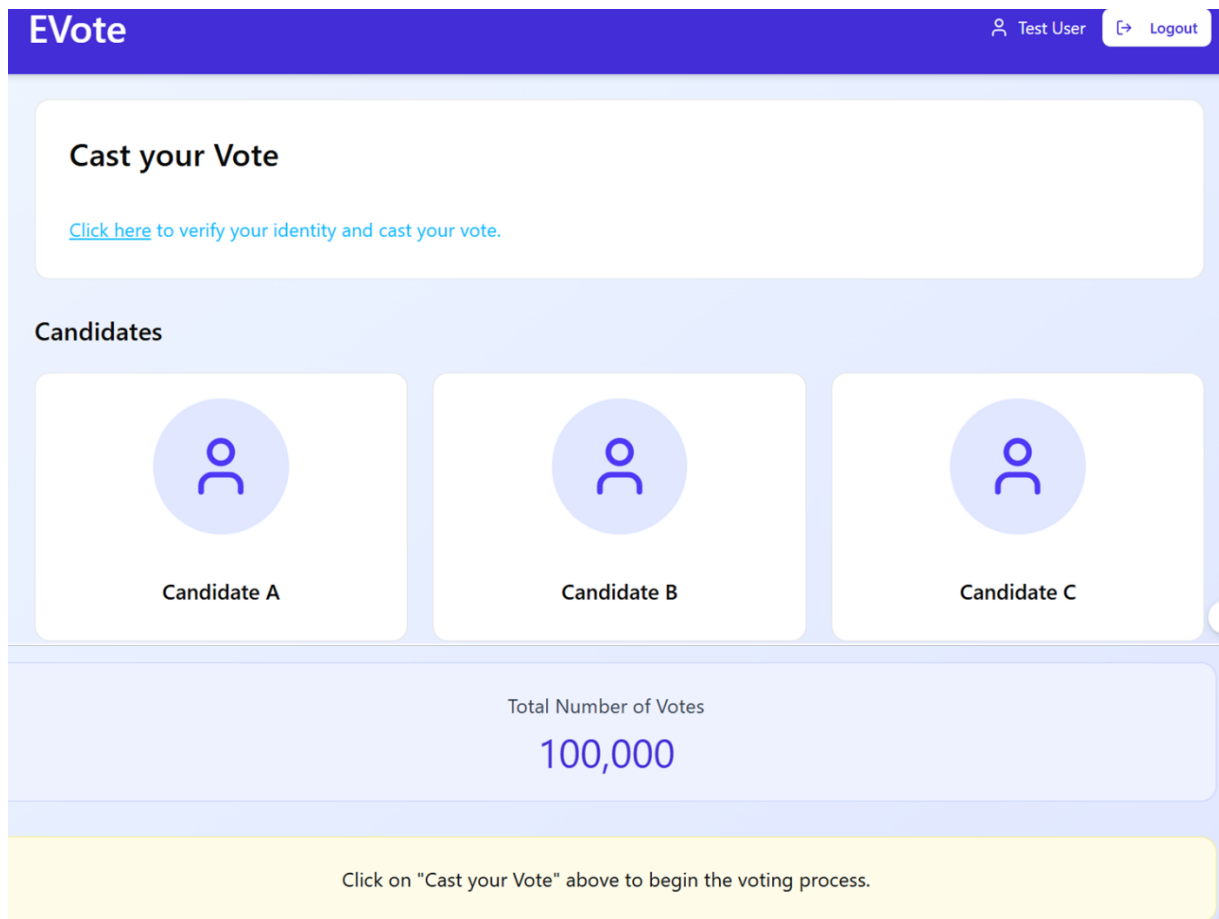


Figure 8: Viewing of Dashboard page of Web Application on Figma

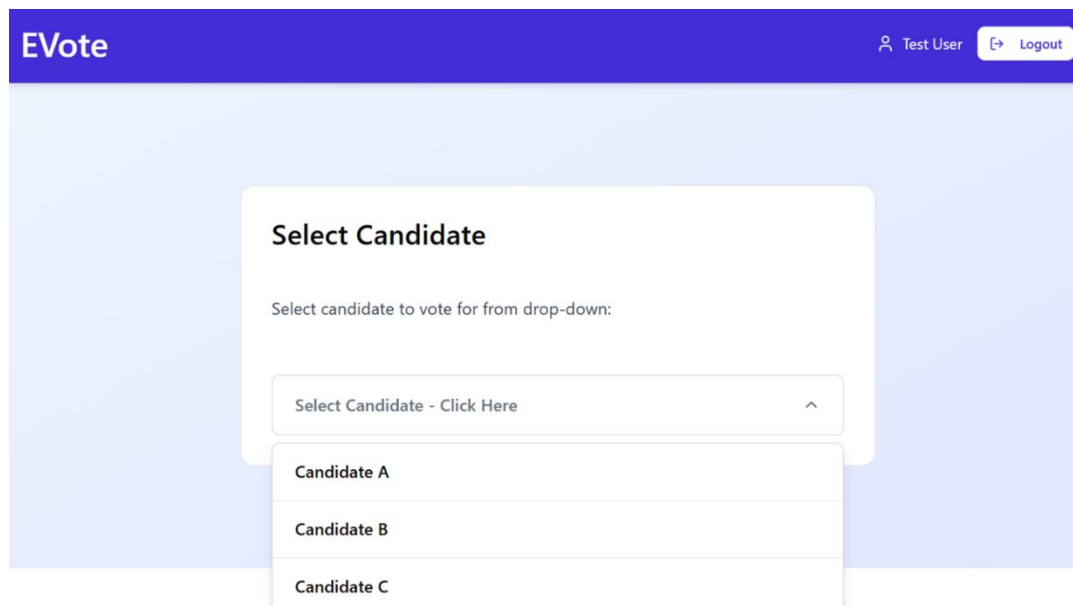


Figure 9: Selection of Candidate page within Web Application on Figma

Select Candidate

Select candidate to vote for from drop-down:

Candidate A



Submit Vote - Click Here

Figure 10: Before submitting vote – confirmation of selection of candidate

Select Candidate

Select candidate to vote for from drop-down:

Candidate A



You have selected **Candidate A**

Please confirm your vote by clicking the button below

Cast Your Vote - Click Here

Figure 11: Confirmation of Candidate vote on figma

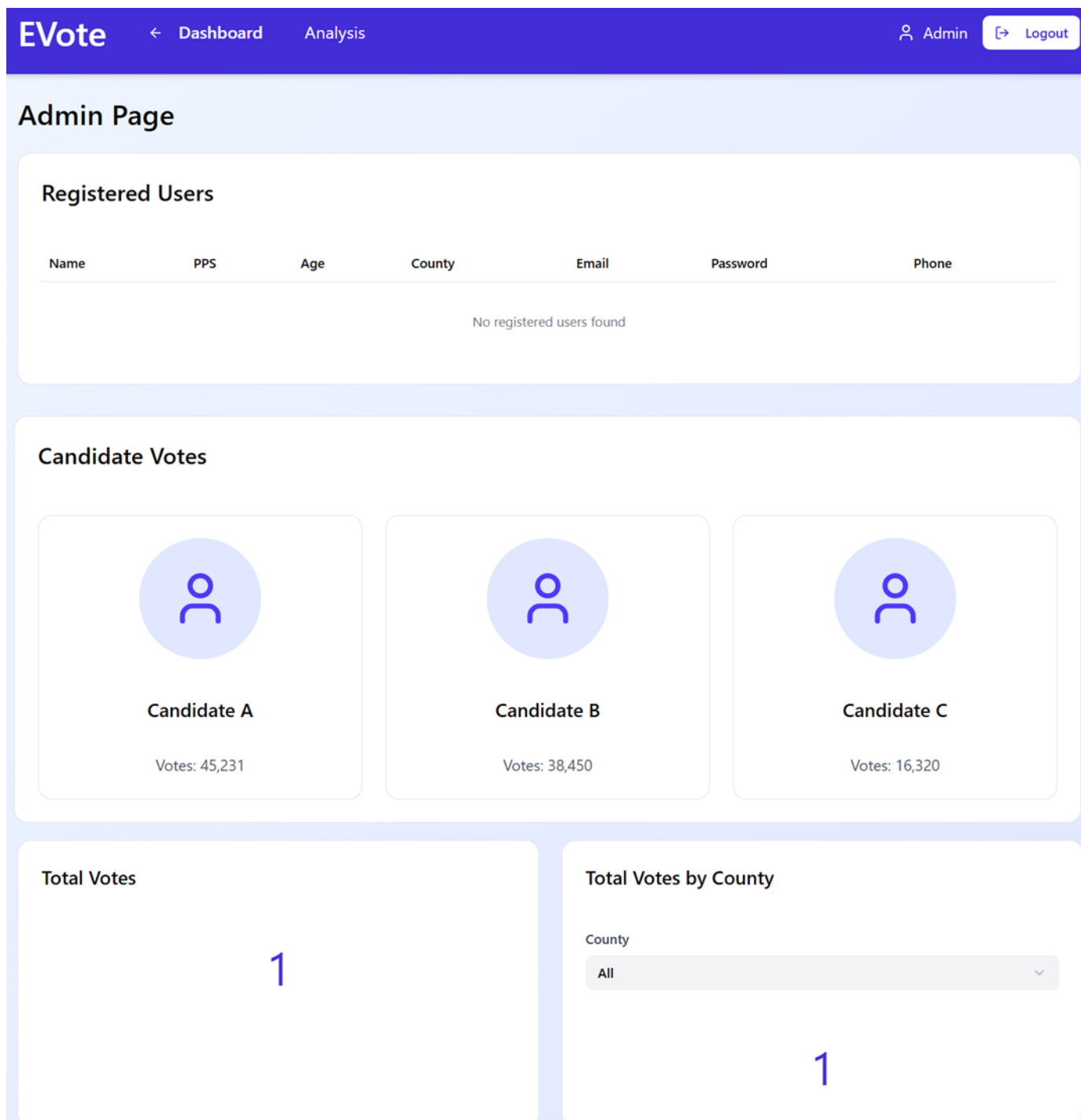


Figure 12: Admin Analysis page of Web Application created on figma

Architecture Diagram

The system being created is better explained using an Architecture diagram. First the diagram was drawn out using a sheet of paper before drawn online (SSW TV | Videos for developers, by developers, 2020).

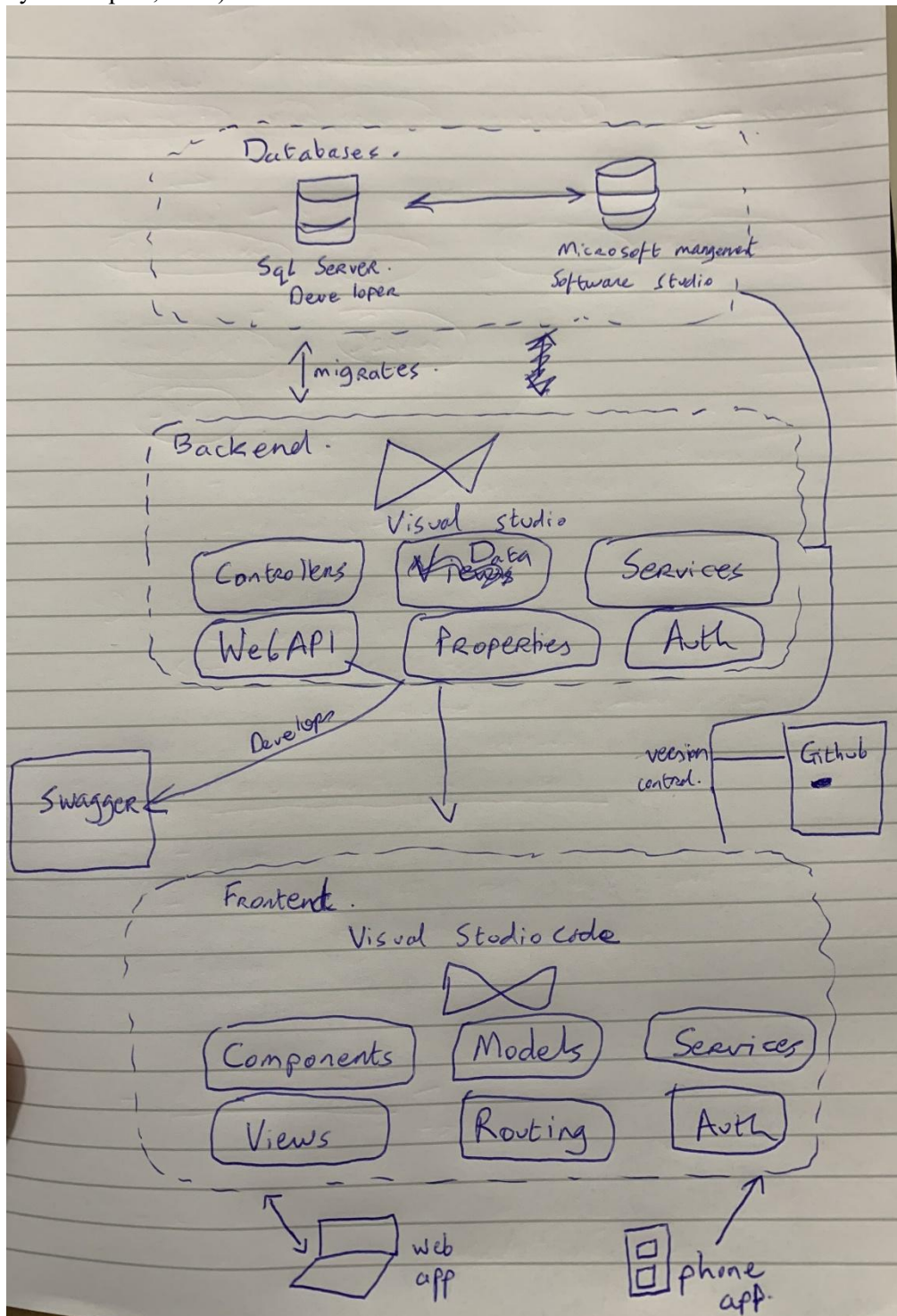


Figure 13: Drawing of architecture diagram

See below the architecture diagram when created using an online platform.

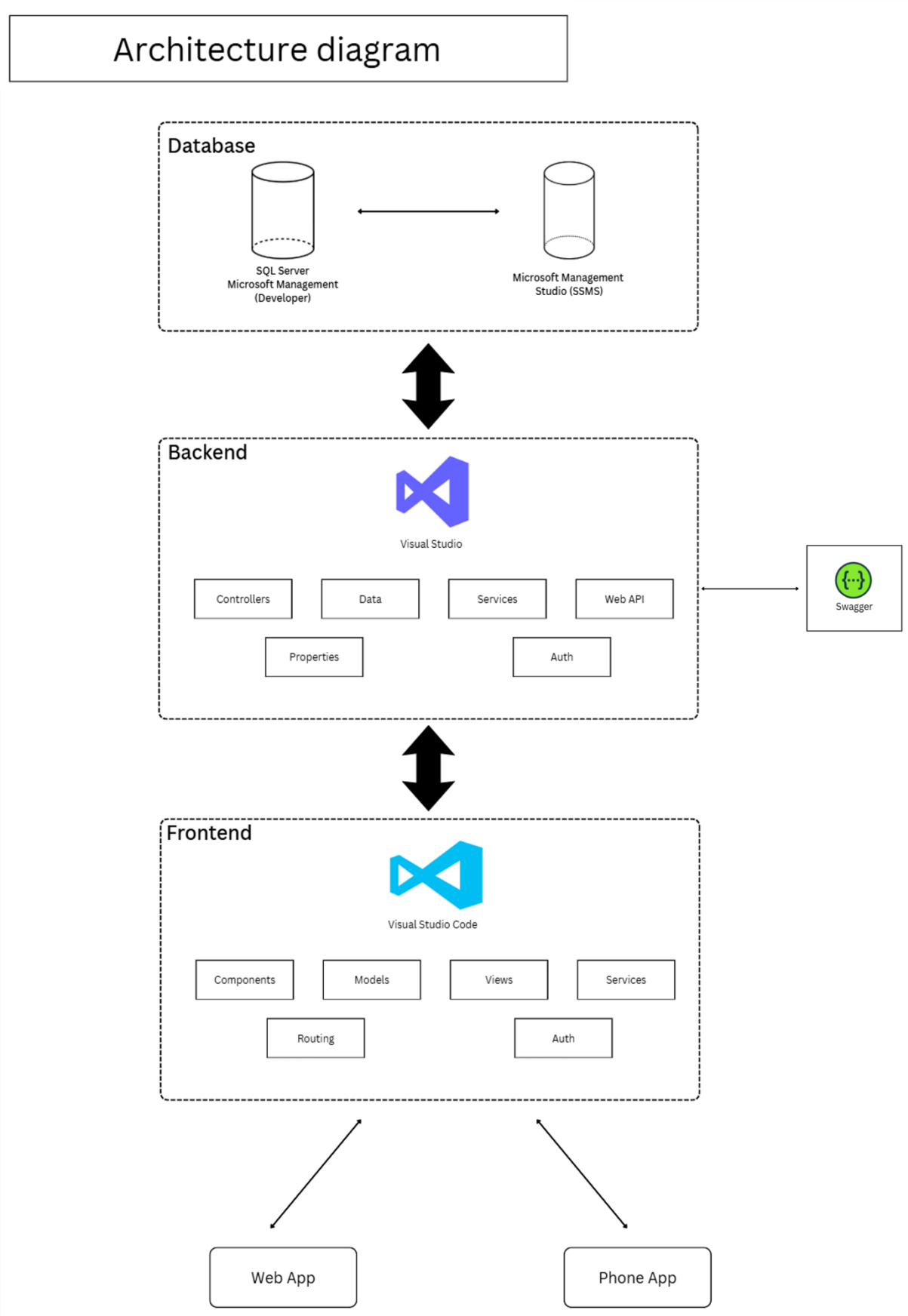


Figure 14: architecture diagram

Architecture Choices

Why I Choose to Use This Architecture Flow

I have been on my work placement over 3 months now. The company I work for are purely Microsoft based and most of the frameworks/languages are brand new concepts to me. Once I knew the flow from SQL Server Database, backend Visual Studio, to Visual Studio Code frontend I was keen to create my own project.

From a personal standpoint I wanted to know to and improve my understanding of my current work placement frameworks, specifically:

- How to create a project using Angular
- How to create a project using .NET
- How to create test projects using .NET
- How use Microsoft PowerShell for CLI commands
- How does a backend web API controller get called from a frontend service
- How the backend connects to the frontend
- How the backend connects to SQL Server database
- How to migrate backend data models to the SQL Server database
- How to update the SQL Server database with new data

Why use Visual Studio

I choose to use Visual Studio for the following reasons:

- To increase my knowledge of Visual Studio as it is a framework I have never used before going on work placement
- The framework Visual Studio is one I need to use often on work placement and will enhance my chances of getting full time employment as now I have created a project with it
- Test my knowledge of .NET
- To get better understanding of layout of the framework, note any shortcuts or tips I come across watching any research videos
- Use a different framework with GitHub and see how I perform
- Gain more insight into the different settings and files needed, that I have not been introduced to yet
- Visual Studio is used backend on my work placement

Why use SQL Server Microsoft Management

I choose to use SQL Server Microsoft Management for the following reasons:

- As built by Microsoft naturally connects to Visual Studio
- Learn a new framework
- Understand how to setup Microsoft's SQL server locally
- Grow a familiarity of how to look at tables, select tables data model, and edit the data within the tables from the SQL Server

Why use Visual Studio Code

I choose Visual Studio Code for the following reasons:

- Already familiar with the framework
- Have connected to GitHub repository before from the framework
- Is used frontend on my work placement

Database Model

A. System Description

Table 4: System Description

Attribute	Action
Admin	Admin manages the user's usernames and passwords
County	A user lives in a town in one of the voting Counties
User	A user provides their details including PPS number, name, address, phone number and email address. A user can also be an admin.
Vote	Once the user's details are provided, they receive one Vote and a registration number for that specific election. A user can log into the website using their registration number and select a candidate.
Candidate	Upon logging into the site, a user will see 3 candidates. Each candidate's name will be displayed along with their campaign slogan. The same candidate can run in other elections should they receive a nomination.
Election	A user will vote in many different elections and in different years. Each time an election is run the number of people who voted will be added to a total count. There are many different types of elections and user will need a registration number and different vote for each one.

Figure 15: Database System Description

B. Enhanced ER Diagram

EVOTE ER Diagram

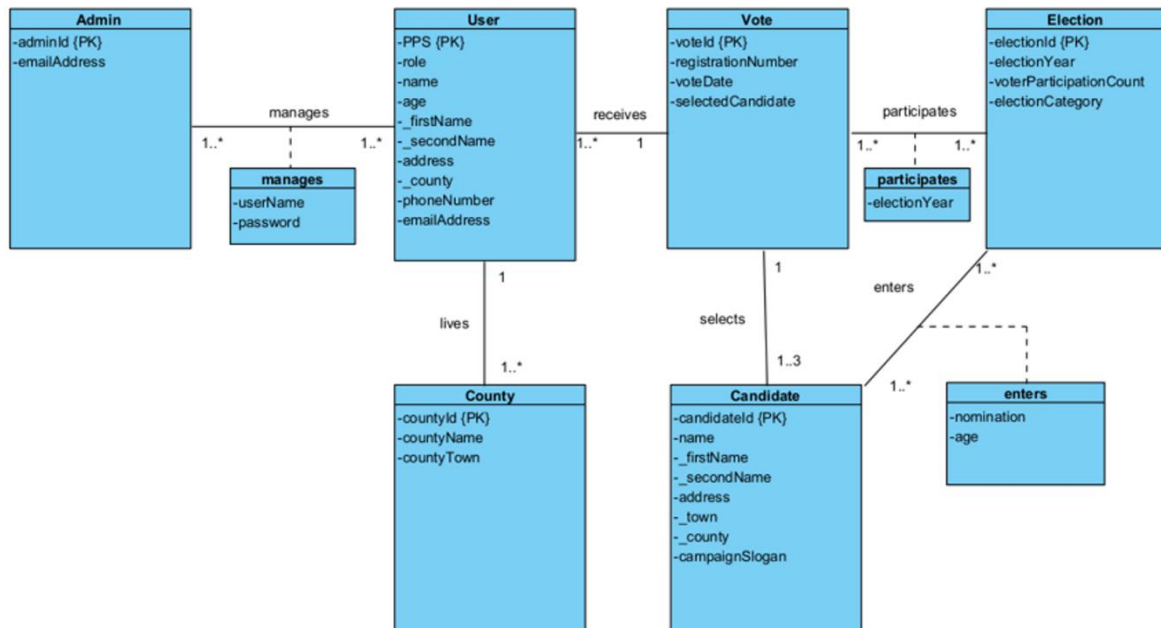


Figure 16: Enhanced ER Diagram

C. Logical Design

Admin (adminId, emailAddress)

Primary key: adminId

Manages (adminId, PPS, userName, password)

Primary Key: adminId, PPS

Foreign key adminId references Admin

Foreign key PPS references User

User (PPS, role, name, age, address, phoneNumber, emailAddress, voteId)

Primary Key: PPS

Foreign key voteId references Vote

County (countyId, countyName, countyTown, PPS)

Primary key: countyId

Foreign key PPS references User

Vote (voteId, registrationNumber, voteDate, selectedCandidate)

Primary key: voteId

Candidate (candidateId, name, address, town, county, campaignSlogan, voteId)

Primary key: candidateId

Foreign key voteId references Vote

Enters (candidateId, electionId, nomination, age)

Primary key: candidateId, electionId, nomination, age

Foreign key candidateId references Candidate

Foreign key electionId references Election

Participates (voteId, electionId, electionYear)

Primary key: voteId, electionId

Foreign key voteId references Vote

Foreign key electionId references Election

Election (electionId, electionYear, voterParticipationCount, electionCategory)

Primary key: electionId

Testing

Swagger will be used for testing on their creation of the API methods. The API methods will be created initially on the service before getting added to a API controller. With Swagger I can view all the API calls and execute each one with mock data to ensure they are functioning correctly.

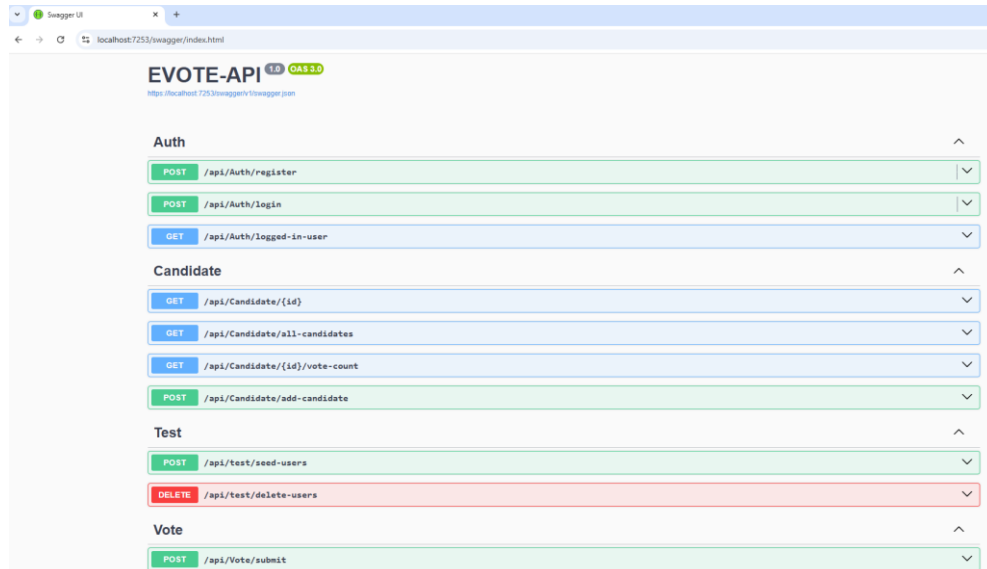


Figure 17: Swagger with all the API Calls from backend

Once I have tested the API calls on Swagger, I can move them to the frontend.

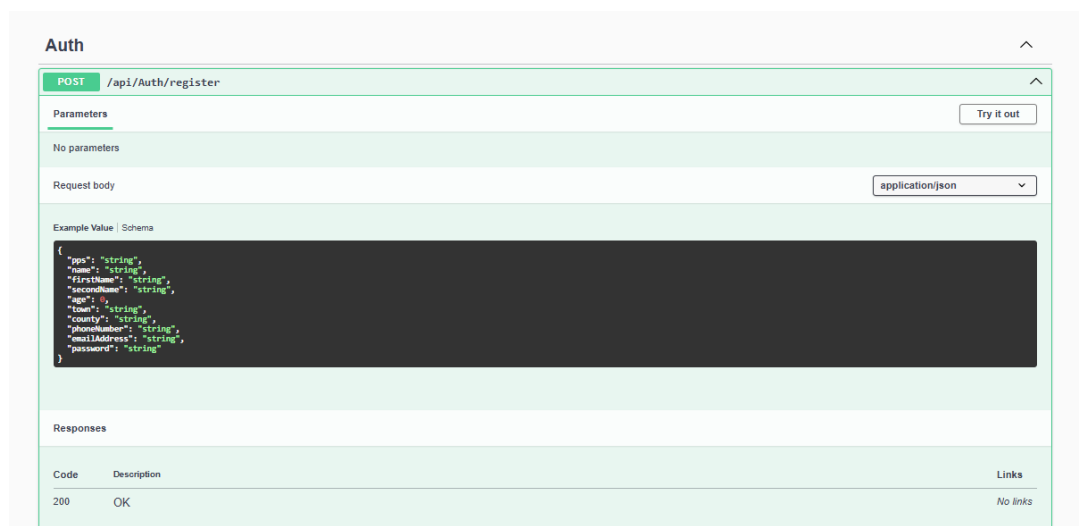


Figure 18: Swagger testing register API method implementation

Unit Tests

Within the backend there will be a test Microsoft Service folder. Each time an API method is created in a controller, another test version will be created in the test folder. Visual Studio gives the user the ability to run tests independently or together. Using this form of Unit tests will allow for the use of mock users, a mock database and to confirm to us that all our controllers are functioning as we would like them to.

Metrics of Success

In order for this project to be successful it needs to meet the following criteria:

- Can a user register
- Can a user login to the voting page using their registered details
- Can a user submit their vote
- Duplicate voting is blocked
- Voting page is blocked unless a user logs in
- There is one or more security features implemented on the project

PART B - Critical Self-Review

What I learned

I learned the following:

- How to develop a Full Stack application
- How to connect Visual Studio to SQL Microsoft Server
- How to create API using .NET
- How to use Angular to develop the frontend
- How to inject dependencies using Angular
- How to develop an application connected to GitHub at the same time. This includes making pull requests, using separate branches and the difficulties that can bring

What I achieved

Time was a large factor in how much work I could get through. With the project being full stack meant that the backend, frontend and database all needed to be setup correctly in order for the application to function properly.

If I had more time, I would have implemented the following:

1. Added more unit tests to the backend Test project
2. Ensured all the database tables got filled with data
3. Added an analysis page which would show all user details for administrators to view
4. Removed the interfaces from the frontend Angular and used classes instead to implement 'cleaner' code usage
5. I would have used `*ngIf equals invalid` and variable name along with the validation message and inserted it into the Template HTML to clean up component element
6. I would have ensured information on the frontend regarding total votes count functions currently when a vote is cast
7. If I had more time I would deploy the application, although this proved difficult as I did try on render initially.

Problems encountered

Some of the problems I encountered while creating the project and how I fixed them were the following:

1. Multiple branches, multiple frameworks – At the beginning of the project I created two branches, ones for the backend and one for the frontend. The issue I found is when I had one framework open, Visual Studio for example, and tried to open Visual Studio Code to work hand in hand with each other, Visual Studio Code would remain on the backend branch. It would empty or worse open my backend folder.

Fix: Create one branch to use on both frameworks

2. Changes to Data Models – Some of my data models changed over time. This meant updating the database.

Fix: Remove the data from the database, create a new migration to the database and update the database.

3. Initially on my backend method to register a user I got back a 400 response from Swagger. It had the message that key size must be greater than '256' bits.

Fix: I realised the secret key I had set did not have enough characters so increased the number of characters and this fixed the issue.

4. I tested my register API method on Swagger knew it was working. I went to test it from end as a user and got blocked by a CORS Policy in the browser.

Fix: Went to the backend and within the root file (program.cs) added a CORS policy that links it <http://localhost:4200> – which is the URL for Angular.

5. Pull Requests that cannot be automatically merged.

Fix: Go through the files and accept or discard a change, then resolve the file and merge the branch.

6. Implementing a project with languages and frameworks I am unfamiliar with Full Stack – I would describe it as more challenging than a problem. However, this is my first ever full stack project and I did using Microsoft frameworks which are not a module in the college. It had its difficulties in terms of understanding how the frontend connects to the backend, and the meaning of code using Angular.

Fix: Watching YouTube demonstration videos and doing courses on Angular including 3 training courses in one week.

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Appendix A

Ethics Checklist

11/29/25, 1:03 PM

Questionnaire Report | SETU Moodle

This Ethics Checklist must be completed for all final year undergraduate, taught postgraduate and research projects in the School of Science and Computing.

View your response(s)

 Respondent: **Jake Nagle** (Group: CM-HDIPCS) Submitted on: Saturday, 29 November 2025, 1:02 PM

Ethics Checklist for Undergraduate, Taught Postgraduate and Research Projects in the School of Science and Computing

All students in the School of Science and Computing who are either (1) in the final year of an undergraduate/BSc degree, or (2) on a taught postgraduate/MSc programme **must complete this Ethics Checklist before conducting their project** regardless of the project type or discipline. The Checklist should also be completed by anyone (whether staff member or student) conducting a **research project** (whether programmatic or not) within the School.

The purpose of this Ethics Checklist is to **identify projects that will require formal ethical approval** from the School Research Ethics Committee, or the SETU Research Ethics Committee, before they can proceed.

Students/applicants should note that this Ethics Checklist is a **formal declaration**, and great care must be taken to **answer all questions accurately**. Students should consult with their project supervisors/advisors regarding any aspects or questions that they are unsure of before completing and submitting the Ethics Checklist.

Students/applicants must **answer all questions** presented to them until the Checklist questionnaire is completed.

?

<https://moodle.setu.ie/mod/questionnaire/myreport.php>

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Feedback Report

No human experimentation issues (UG).

No animal experimentation issues (N/A).

No issues regarding the use of human tissues.

No animal tissue or biological fluids issues.

No ionising radiation issues.

No primary data collection issues (N/A).

<https://moodle.setu.ie/mod/questionnaire/myreport.php>

2/8

No underage/vulnerable people issues (UG).

No issues regarding existing/secondary data use (N/A).

No controversial data issues.

No issues related to the collection of rare or protected plants.

No issues regarding the use of genetically modified (GM) plant material.

Instructions:

1. If the above feedback is **entirely green** then, based on your answers, there is **no need to apply for ethical approval** for your project.
2. If **any** part of the above feedback is **yellow/amber**, then there is at least one issue with your project that needs to be reviewed and **you must apply for ethical approval** to continue your project.
3. If **any** part of the above feedback is **red** then there is a serious ethical issue and **you cannot continue your project** as currently planned.

<https://moodle.setu.ie/mod/questionnaire/myreport.php>

3/8

It is recommended that you print this Feedback Report to a PDF file for your records. You should also forward and discuss this Feedback Report PDF with your project supervisor. They will be able to advise if you have any further questions or if you need to apply for ethical approval.

Please review this report carefully to make sure that all your answers have been properly recorded.

Note that you can ignore all "white exclamation marks in red circles" in this document

1 Are you a student on a **final year undergraduate** programme, a **taught postgraduate** programme, or are you conducting a **research project**?

- ☒ Final Year Undergraduate
☐ Taught Postgraduate
☐ Postgraduate Research Project
☐ Other Research Project

2 What is the **working title** of your project?

EVOTE – An imaginary secure, user friendly electronic voting system for election polling implemented as a full stack application addressing UI for disabilities (accessibility), spoiled votes, including a dashboard for results.

3 Who are the project **supervisors/advisors/principal investigators**?

Colm Dunphy / TBC

4 Does your project involve **human experimentation**?

- ☐ Yes ☒ No

5 Does your project involve **live animal experimentation**?

- ☐ Yes ☒ No

<https://moodle.setu.ie/mod/questionnaire/myreport.php>

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(6) Is the planned animal experimentation limited to **non-invasive procedures only** (such as feeding, weighing, or taking naturally voided faecal or hair samples), and does **not** involve any invasive procedures (such as taking rectal faecal samples or blood) from live animals?

- ☐ Yes ☒ No

7 Does your project involve the use of **human remains/cadavers/tissues/cells/biological fluids/embryos/foetuses**?

- ☐ Yes ☒ No

(8) Do you intend to only use established **commercial human cell lines**, and no other **human remains/cadavers/tissues/cells/biological fluids/embryos/foetuses** in your project?

- ☐ Yes ☒ No

9 Does your project involve the use of **animal cells, tissues or biological fluids**?

- ☐ Yes ☒ No

(10) Do you intend to only use (1) **established commercial animal cell lines**, or (2) **slaughterhouse-derived tissues/fluids**, or (3) **fluids collected as part of routine animal husbandry** (e.g. milk) and no other animal tissues or biological fluids in your project?

- ☐ Yes ☒ No

11 Does your project involve the **collection of rare or protected plants**?

- ☐ Yes ☒ No

12 Does your project involve the generation or use of **genetically modified (GM) plant material**?

- ☐ Yes ☒ No

<https://moodle.setu.ie/mod/questionnaire/myreport.php>

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- (13)** Do you agree to (1) only use **established genetically modified (GM) plant cell lines, seeds, or plant products** in your project, (2) **not generate new plant mutations** using chemical or other means, and (3) follow specified SETU **containment and use protocols** for GM plant materials at all times?

☐ Yes ☐ No

- 14** Does your project involve the use of **ionising radiation**? (e.g. use of gamma ray spectrometry)

☐ Yes ☒ No

- (15)** Do you agree to carefully **follow the instructions** of the SETU designated **Radiation Protection Officer (RPO)**, and **adhere to all legal requirements** as set out in the Radiological Protection Act 1991 (Ionising Radiation) Regulations (2019), regarding the use of ionising radiation materials and equipment?

☐ Yes ☐ No

- 16** Does your project involve the **collection of any new (or primary) data** from **individual people or groups**?

☐ Yes ☒ No

- (17)** Does your project involve the **collection of any new (or primary) individual or group data** that is **personally or uniquely identifying**? (e.g. data about people or organisations/companies/groups that could be used to identify those individuals or groups; data collection might take any form, including internet and social media data, etc.)

☐ Yes ☐ No

- (18)** Will you ensure that participants who you are collecting data from are provided with **fair warning** and must provide **explicit informed consent** for any data collected?

☐ Yes ☐ No

- (19)** Will you ensure that any project-related data collection, data storage, and data use is in **full compliance** with the EU **General Data Protection Regulation (GDPR)** and the **Data Protection Act (2018)**?

☐ Yes ☐ No

- (20)** Does any of the data that you intend to collect include **sensitive or private personal information** about individuals, or **commercially sensitive information** about organisations/companies/groups?

☐ Yes ☐ No

- 21** Does your project involve **persons under the age of 18 years** (i.e. minors), or **any vulnerable groups**? (e.g. prisoners, refugees, those in care, addiction service users, etc.)

☐ Yes ☒ No

- 22** Does your project involve the use of **existing (or secondary) human data**? (i.e. data originally collected for another purpose)

☐ Yes ☒ No

- (23)** Is the existing or secondary human data you intend to use either (1) **anonymous/non-personally identifying** and in the **public domain**, or (2) available with **explicit and specific informed consent or permission** for the data to be **legally** reused in the way you intend?

☐ Yes ☐ No

- (24)** Are any aspects of the primary/secondary data you intend to use for the project **controversial** in nature?

☐ Yes ☐ No

Before you submit the Ethics Checklist, you must **confirm all of the following**:

25



- ☒ I understand that the Ethics Checklist is a formal declaration.
- ☒ I have answered all questions on the Ethics Checklist carefully and truthfully.
- ☒ The supervisor/advisor (or principal investigator) for the project is present as the Ethics Checklist is being submitted, or they have given me explicit permission to submit it in their absence.
- ☒ I have had adequate ethics training and/or instruction prior to completing the Ethics Checklist.
- ☒ I understand, and agree to abide by, the general ethical principle of "do no harm" for this project.
- ☒ I will follow the instructions given in the Feedback Report.

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Authentication Code (ask your project supervisor/advisor for this code)

Enter Student Number: Enter the Authentication Code below and click "Verify Code" **Note: If an INVALID authentication code is used then this submission is NULL and VOID**

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Appendix B

Client side Paper drawings – designing the Frontend

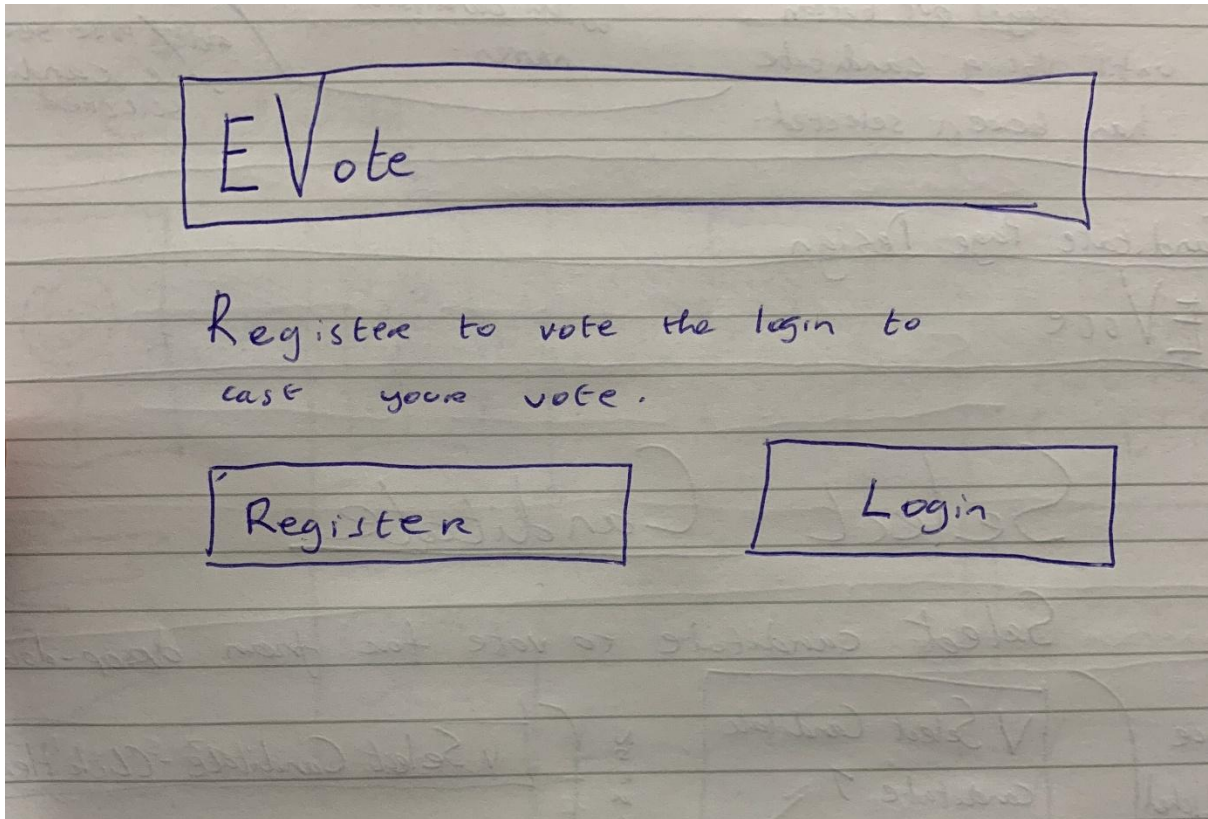


Figure 2.0 Drawing of home page of Web Application

Login Page.

EVote

Enter your detail to login and Vote.

PPS/Email:

<place holder text />

EVote Password.

Note: You must register before you login and vote. If you have not do so register here

Figure 2.2 Drawing of login page of Web Application

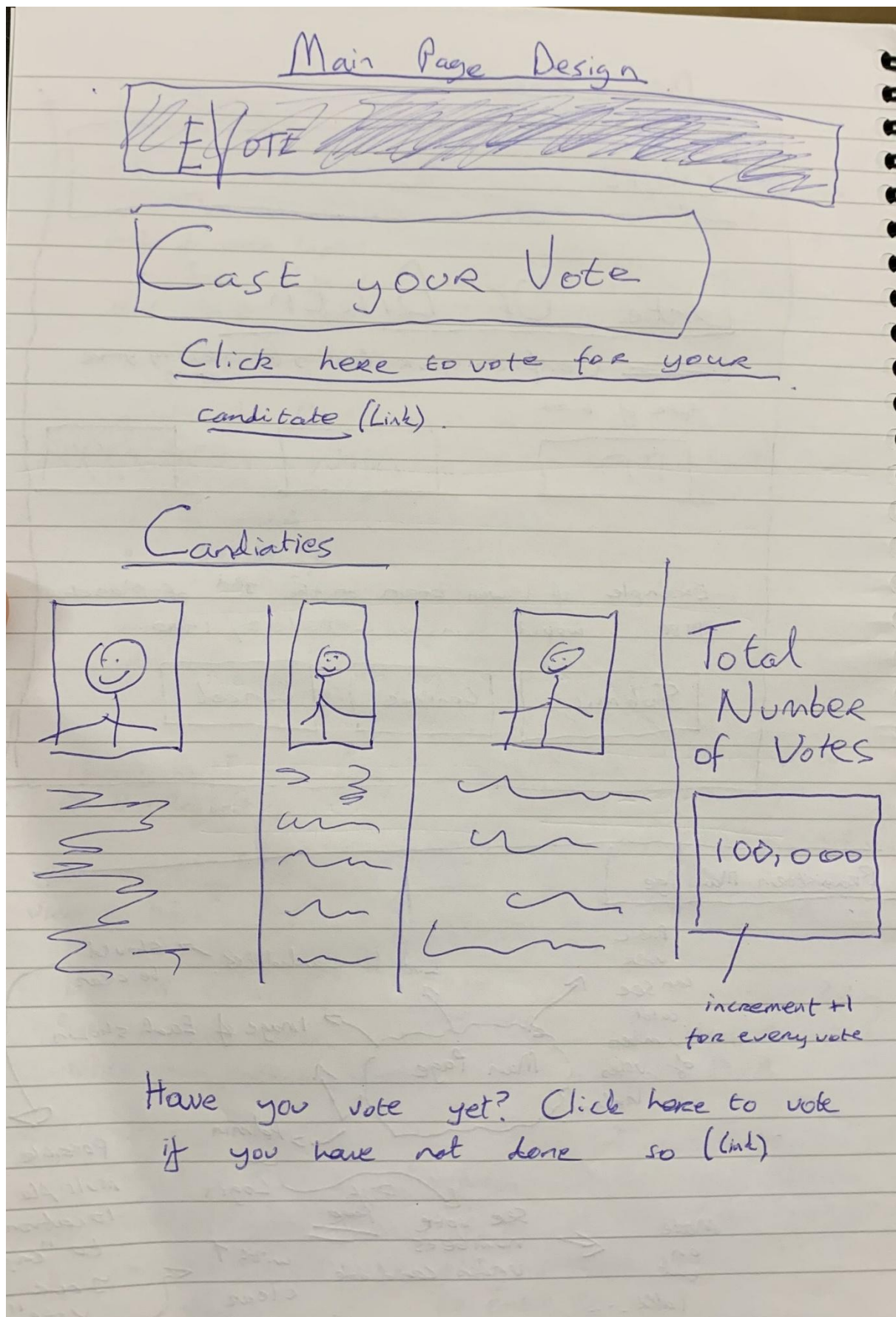


Figure 2.5 Drawing of login page of Web Application

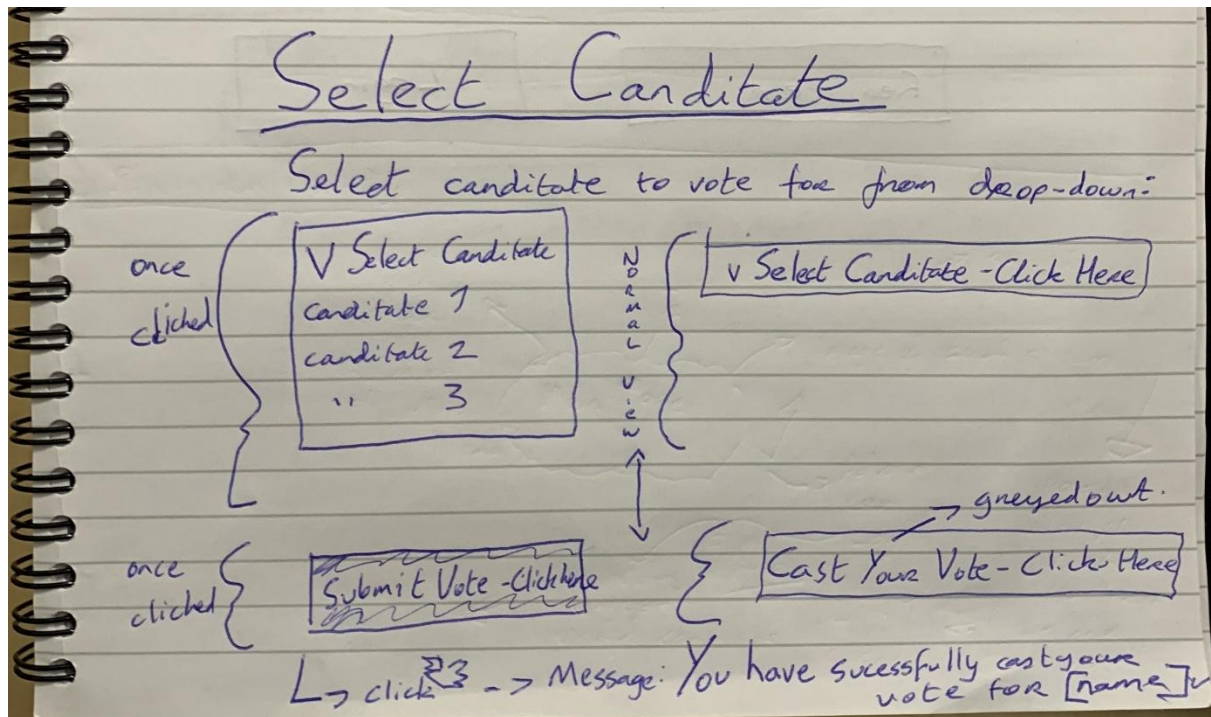


Figure 2.7 Drawing of Voting page of Web Application

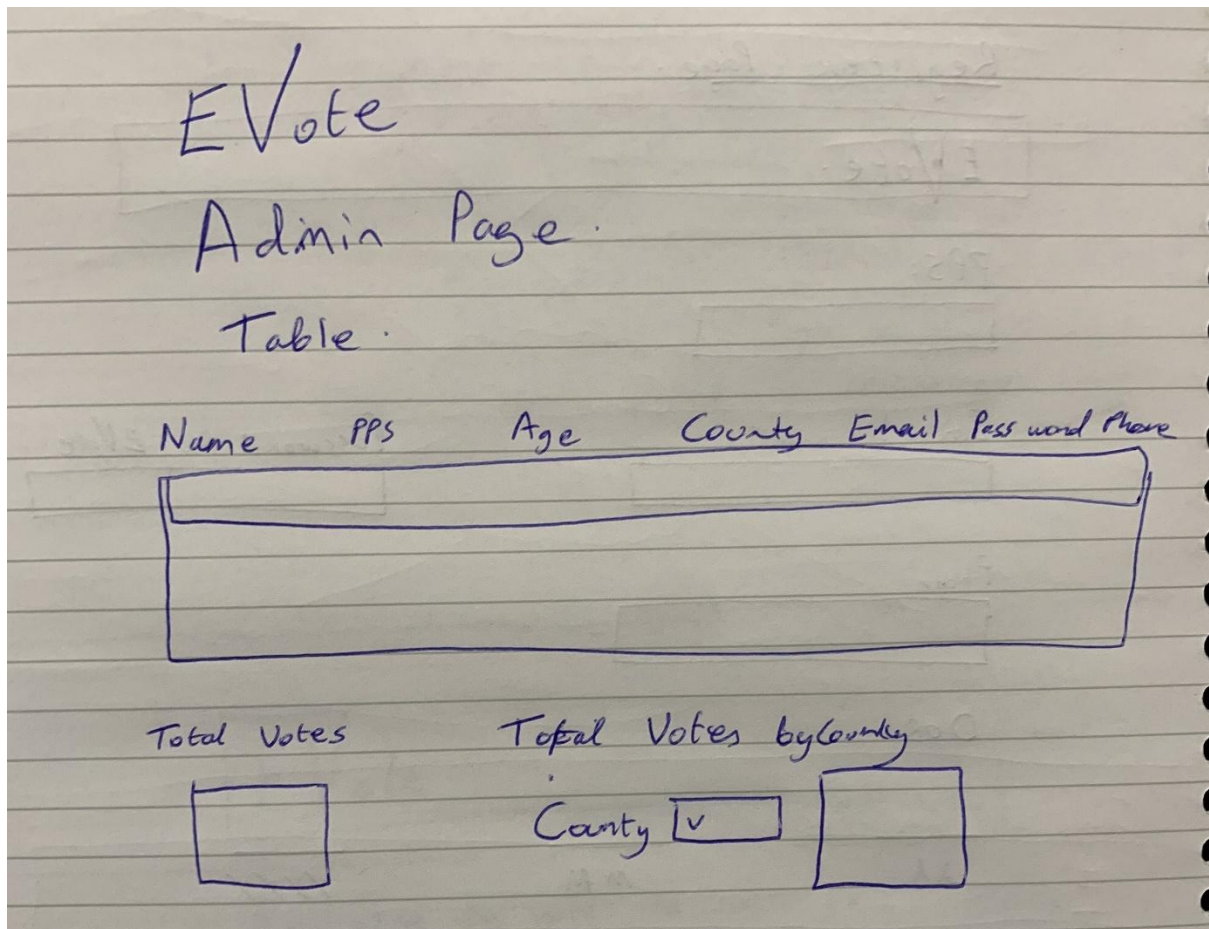


Figure 3.0 Drawing of Admin Analysis page of Web Application